



BHUVNESH KUMAR



ACADEMIC DETAILS

Year	Degree / Board	Institute	GPA / Marks(%)
---	M.Tech in Computer Science & Engineering	Indian Institute of Technology Delhi	8.47
2019	B.Tech in Computer Science & Engineering	ABES Institute of Technology, Gzb	69.52 %
2015	12th (CBSE)	S.C. DAV Public School, Noida	82.80 %
2013	10th (CBSE)	Sahil Public School, Noida	8.2

WORK EXPERIENCE

- **Sopra Steria, Noida (Aeroline Team) | Full Stack Developer** (Aug, 2019 - Feb, 2021)
 - **Project: BFEMS** (Buyer Furnished Equipment Management System): An AIRBUS web application used for managing the equipment required for aircraft assembly.
 - A legacy project built on **PL/SQL** (Oracle Database) and **.NET MVC**, interfacing with multiple applications that provide/require large sets of aircraft and equipment data.
 - Solely handled this critical application of AIRBUS, ensuring continuous availability, reliability & smooth functionality.
 - **Role:** Involved in Designing, Development and Testing of Web Application.
 - Additionally, completed a **6-month** internship with the same organization, contributing to the same project.
 - Honoured with Falcon award for **quality delivery** of v12 release of application.

IIT DELHI THESIS

- Title:** Fake Base Station Detection At Network Side (Thesis Defended)
- Supervisor:** Prof. Vireshwar Kumar
- Description:** Designed a machine learning based detection framework to detect fake base stations (FBS) in 5G network.
- Built a comprehensive **NS-3 (5G-LENA)** simulation of a dense **urban microcell** network (12 gNBs, 300 UEs) to model mobility and voice traffic scenarios.
 - Implemented a **regression-clustering pipeline** (Random Forest + KMeans) to detect anomalies in RSRP values.
 - Validated detection under diverse configurations by **varying transmission power levels** of legitimate and fake gNBs.
 - Also enhanced the **srsRAN 5G UE** prototype to support measurement reporting (periodic + event-triggered A1-A6) feature according to **3GPP TS 38.331** standards.

PROJECTS

- **Enhancement in Xv6 Operating System (Prof. Abhilash Jindal) :** (Jan, 2024 - Apr, 2024)
 - Implemented **write-ahead** logging using **undo-logs**, which helps to ensure crash consistency.
 - Designed a **process scheduler** with a customised round-robin scheduling policy, prioritising foreground processes over background processes based on time-sharing criteria.
 - Added **page-swapping** with **copy-on-write** mechanism. Modified the disk layout to include swap space and implemented a page fault handling logic that enables sharing COW pages between forked processes.
- **Contra Game Development from Scratch (Prof. Smruti Ranjan Sarangi) :** (Sept, 2023 - Oct, 2023)
 - Designed and developed a 2D platformer game with 3 levels in **Unity** using **C#** language.
 - Key features: Player controls, health, Enemies, Boss battle, Power-ups, traps.
- **Markdown to HTML Converter (Prof. Smruti Ranjan Sarangi) :** (Jul, 2023 - Aug, 2023)
 - Built a compiler using **Flex** for lexical analysis and **Bison** for parsing to generate **HTML** from **Markdown**.
- **Trivial File Transfer Protocol (Prof. Smruti Ranjan Sarangi) :** (Oct, 2023 - Nov, 2023)
 - Implemented a TFTP server and a client from scratch in **C++** in compliance with RFC 1350.
 - Integrated with file compression/decompression through Huffman coding and custom encryption/decryption.

POSITIONS OF RESPONSIBILITY

- **System Administrator, CSE Department (Supervisor: Prof. Abhilash Jindal):** (Aug, 2023 - till date)
 - Ensuring **reliable system performance & availability** by proactively **monitoring IT resources & resolving issues**.
 - Deployed & maintaining an **e-help ticketing system** for CSE department.
 - Deployed a **Room booking system** for CSE department.
 - Deployed & maintaining an **independent Moodle server** for CSE department with **8 VPL-jail** servers to ensure reliable resource management.
 - Administered the Moodle server during **4 departmental entrance exams**, ensuring zero downtime for all candidates.

TECHNICAL SKILLS

- **Languages:** C/C++, Java, C#, Python, SQL, PL/SQL, Shell, Pro*C
- **Databases:** Oracle SQL Dev., MySQL
- **Framework/Tools:** .NET, Docker, Unity, Git
- **Modelling Language:** UML (Use Case Diagrams)



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IIT COURSE

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COURSES DONE

Advanced Data Structures, Software Systems Laboratory, Resource Management In Computer Systems, Minor Project, Computer Networks, M.tech Project Parti, M.tech Project Partii, Research Practicum In Computer Science And Engineering